

Cryptchat

Problem Statement

Journalists, politicians, and activists face increasing surveillance and censorship when communicating sensitive information. Traditional encrypted messaging platforms:

- Are often blocked or monitored
- Clearly indicate secure communications are occurring
- Leave metadata trails that can compromise users

There is a critical need for a communication tool that:

1. Provides strong encryption
2. Conceals the fact communication is happening
3. Works through standard, unblocked platforms
4. Offers plausible deniability

Our Solution: CryptChat

CryptChat solves these challenges by combining:

- **Military-grade encryption** (multiple algorithm options)
- **Advanced steganography** (hiding messages in images)
- **Platform-agnostic delivery** (works through any image-sharing service)

Key Innovations

1. **Selectable Encryption:**
 - AES-256-GCM (government standard)
 - ChaCha20-Poly1305 (mobile optimized)
 - RSA-2048 + AES (asymmetric option)
2. **Adaptive Steganography:**
 - LSB encoding for PNGs (high capacity)
 - DCT coefficients for JPEGs (social media friendly)
 - Automatic format selection
3. **Covert Delivery:**
 - Messages hidden in ordinary-looking images
 - Shareable via any platform (WhatsApp, telegram, email, social media)
 - No visible indication of secure content

Implementation Overview

Technical Architecture

1. **Encryption Layer:**
 - User-selectable algorithms
 - Secure key generation and storage
 - PBKDF2 password hardening

Algorithm	Pros	Use Case
AES-256-GCM	Fast, widely audited, FIPS-approved.	Best for most users.
ChaCha20-Poly1305	Faster on mobile, resistant to side-channel attacks.	Ideal for older devices.
RSA-2048 + AES	Asymmetric (no pre-shared key needed).	For advanced users.

2. **Steganography Engine:**
 - Lossless embedding for PNG
 - Compression-resistant JPEG hiding
 - Automatic capacity optimization
3. **User Interface:**
 - Simple 3-step workflow:
 1. Write message + select image
 2. Choose encryption method
 3. Share disguised image
 - Intuitive decryption flow

Security Features

- Zero-knowledge design (we can't access your data)
- Optional self-destructing messages (in further stages of development)
- Metadata scrubbing
- Anti-forensic techniques

Why CryptChat Stands Out

Unlike traditional secure messengers:

- ✓ Messages don't appear as "secure" communications
- ✓ Works through channels already allowed in restrictive environments
- ✓ Provides multiple layers of deniability
- ✓ Adapts to user's technical level

Roles

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